Homework 2

First:

Be sure to use a .qmd file for this assignment.

Submit your .qmd file, as well as the rendered .html or .pdf file. If you render to .html, please be sure to `embed-resources` in your qmd YAML (see below). If you render to .pdf, you do NOT need to do this.

```
format:
  html:
    embed-resources: true
```

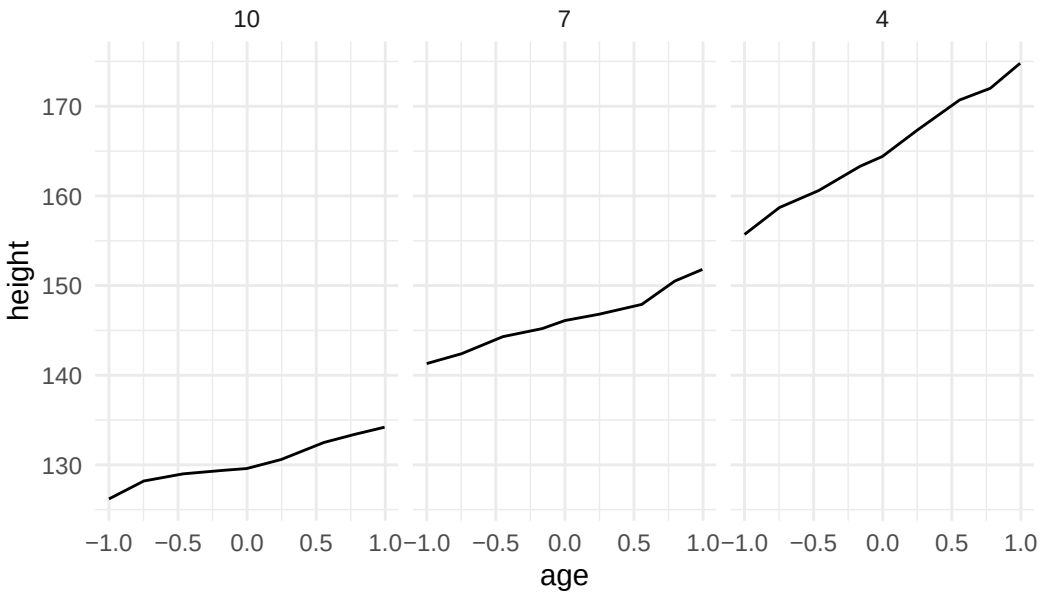
Then:

1. Install the `{nlme}` package
2. Load `{nlme}`, `{janitor}`, and the `{tidyverse}` packages
3. Run the chunk below to prep some data from the `{nmle}` package for plotting

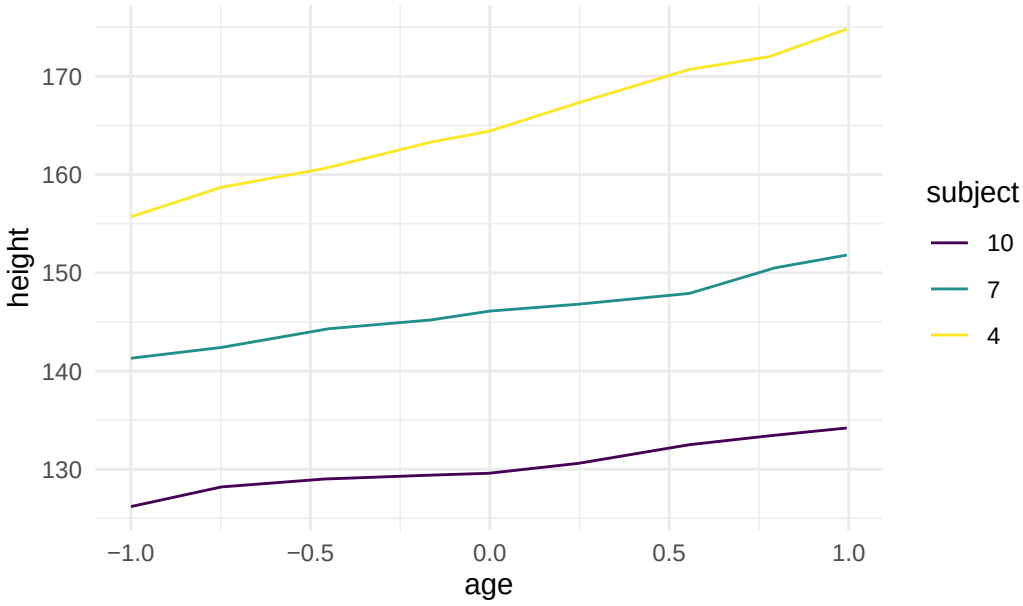
```
pd <- Oxboys %>%
  clean_names() %>%
  mutate(subject = factor(subject),
         occasion = factor(occasion)) %>%
  filter(subject == "10" | subject == "4" | subject == "7") %>%
  as_tibble()
```

4. Reproduce the following two plots, using the *pd* data. You can use whatever theme you want (I used `theme_minimal()`), but all else should be the same. Note that Plot 1 is a line plot, not a "smooth." Note that Plot 2 is a line plot also.

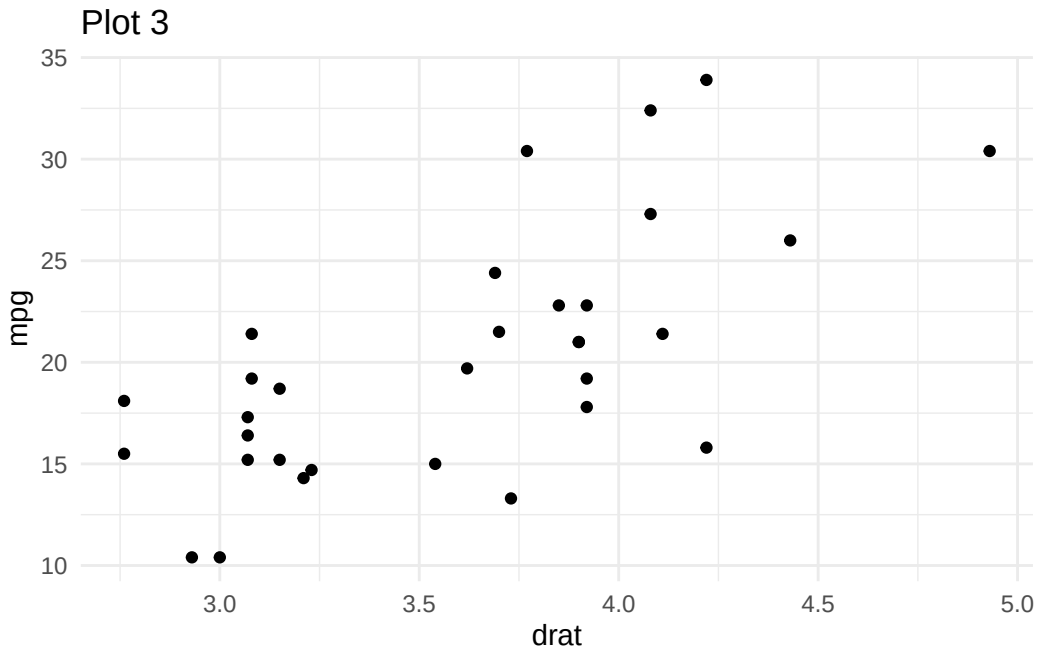
Plot 1

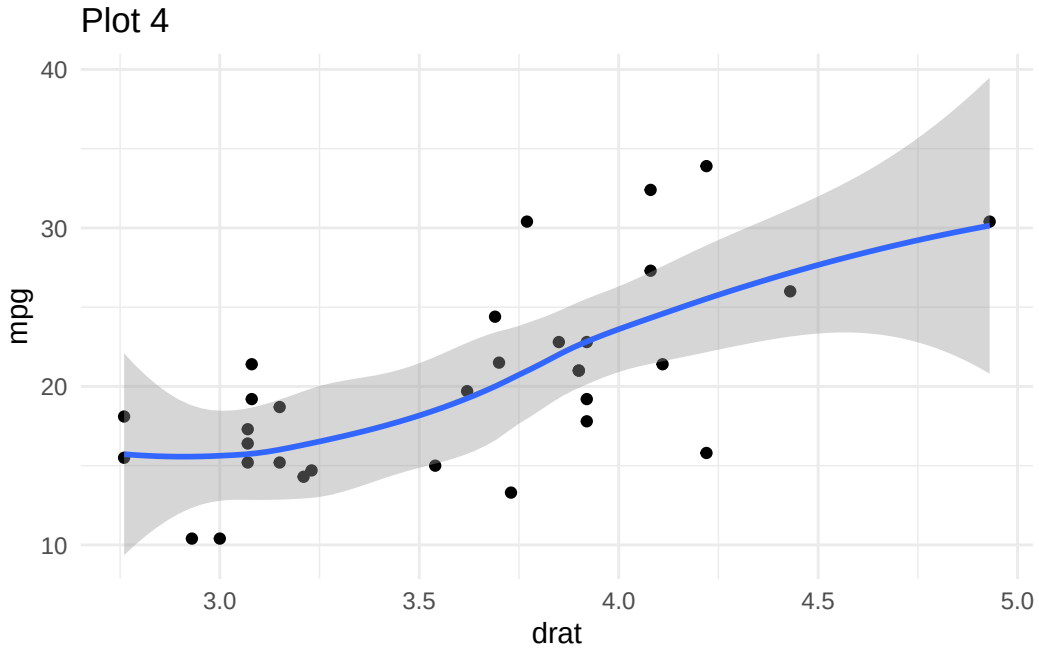


Plot 2

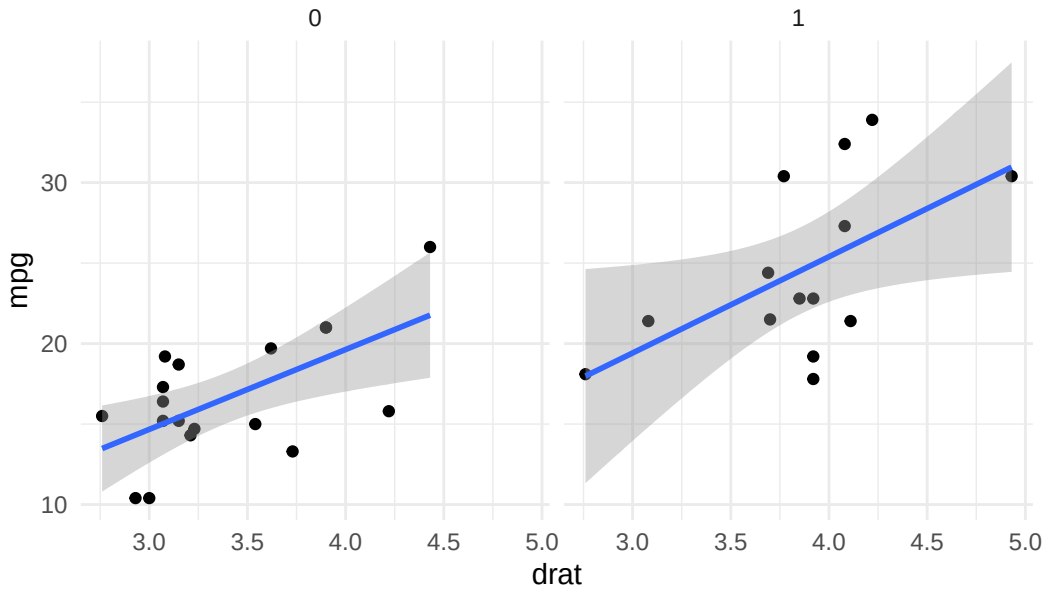


5. Use the *mtcars* dataset from base `R` to replicate the following plots. (Just type *mtcars* into the console to see the dataset).

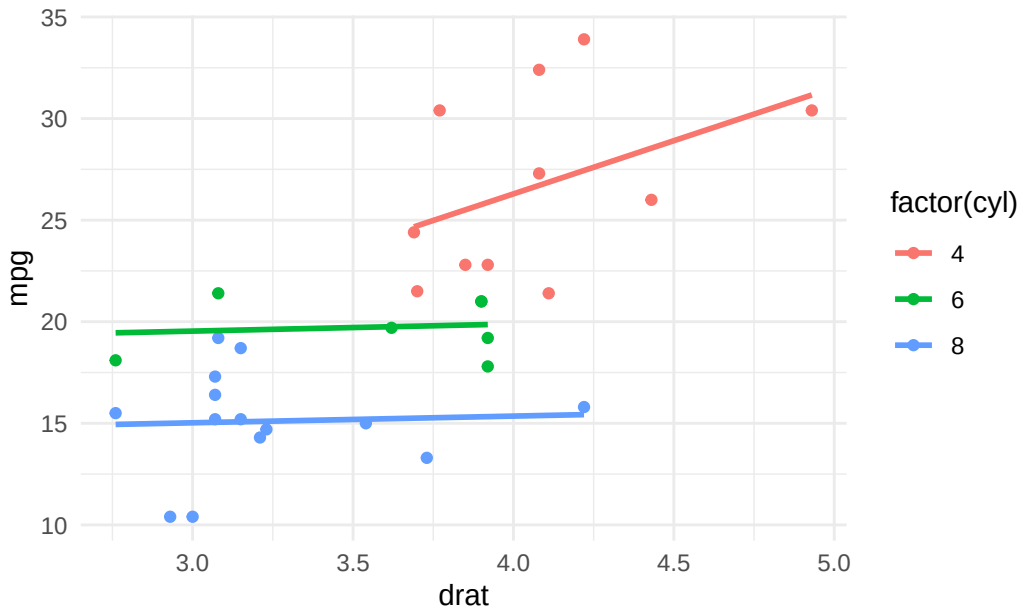




Plot 5


0
1
mpg
30
20
10
3.0
3.5
4.0
4.5
5.0
drat

Plot 6


35
30
25
20
15
10
mpg
3.0
3.5
4.0
4.5
5.0
drat
factor(cyl)
4
6
8

6. Use the *Orange* dataset, also part of base `R`, to replicate the following plots. See slides from the week 3 class for labels.

